Computer Organization and Architecture
Assignment-1

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Submitted To:

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1. What is Yotta?

Yotta is an Indian Company that offers services for cloud computing and services for data center.

Yotta Features:-

- Services of cloud computing
- Security of data stored
- Less use of energy in data centers
- Options for recovery at time of disaster

Uses:-

- Artificial Intelligence (AI)
- Medical care
- Services for banking
- Public services
- Education and research

2. What is NPTEL?

NPTEL or the National Programme on Technology Enhanced Learning is a project that is funded by the Ministry of Education,Indian Government, to provide certification and online courses in various areas.

It provides free online courses, video lectures by professors from IIT, and certificates after passing the exam.

It increases the awareness of technology, helps us in placements and internships, gives industry known certificates and the best part is we can learn at any place and at any time.

NPTEL is a very good platform for students to boost their skills and learn from the good educational institutions of India.

3. Who won Turing Award recently? What is the Turing Award?

The A.M. Turing Award is the highest distinction in computer science and is often called the “Nobel Prize of Computing.” It honours those who have contributed significantly to the area of Science and computing.

The most recent award was given in 2026 to Andrew G. Barto and Richard S. Sutton. The prize was for their work in Reinforcement Learning, a major branch of Artificial Intelligence. Their work helped lay the foundation for how computers learn from experience, experimenting with different actions, getting feedback in the form of rewards or penalties, and getting better at a task over time. Their work has had a great impact on the way machines learn and make decisions, from robotics to game-playing AI.

4. What is Raspberry-Pi ?

Raspberry Pi is a small and less expensive computer designed by the Raspberry Pi Foundation in the United Kingdom. It was released in 2012 to help students learn programming, computer science and electronics in an easy and practical way. It’s about the size of a credit card but can do many of the things a normal computer can do such as browsing the internet, playing videos, writing programs and running an operating system.

Raspberry Pi is used at large scale in robotics, IoT, home automation, AI and embedded system projects.

It is a low cost mini computer with the help of which students build innovative projects.

5. What are the parameters to decide which programming language should be used to develop an application?

The choice of programming language depends on the following parameters :-

- type of application
- required speed and safety
- platform support
- simplicity of development

For instance, Python is preferred for AI, Java in case of Android apps, and C++ for system software and games. Choosing the right language is a way to increase the performance and reduce the development time.

6. Why is GPU faster than CPU ?

A Graphics Processing Unit (GPU) is faster than a Central Processing Unit (CPU) at things that need lots of similar calculations . That's because a GPU has hundreds or thousands of little processing cores that can work on many tasks at once. This ability is called parallel processing and is considered very good for applications like gaming, graphics rendering, artificial intelligence and video editing. A CPU , however , has less cores , but they are more powerful and are designed for different types of tasks one at a time . So,CPU is better for general computing while GPU is better for heavy computational work that can be done at the same time.

7. Explain CPU,GPU,TPU.

CPU	GPU	TPU
It means Central Processing Unit.	The abbreviation means Graphics Processing Unit.	It stands for Tensor Processing Unit.
Performs general computer work.	Graphics and parallel-processing focused.	Made specifically for Artificial Intelligence.
Has few powerful nucleus.	Has a lot of smaller cores.	Uses AI processing units.
For your everyday computing.	Used in graphics and gaming.	For use in machine learning and deep learning.

8. RAM Size,Word Size,Speed and Operating System of my Computer:-

Lenovo Intel Core i7
Operating System:-Microsoft Windows 11
RAM Size:-32.0 GB
Speed:-5600 MT/s(MT=Mega Transfers)
Word Size:-64-bit(x64-based)

9. Who is the founder of Intel,AMD?
Intel was founded by Gordon Moore and Robert Noyce.
Jerry Sanders and seven co-founders founded AMD (Advanced Micro Devices) .

10. What is the CPU organization of the processor used in our laptop?
My laptop has an Intel Core i7 processor. The Core i7 is based on the x64 (64-bit) architecture.

11. What is the Stampeding Herd Problem?
The Stampeding Herd Problem occurs when multiple processes or threads are waiting on the same resource. And when that resource is available, they all try to tap that resource at once. All run together. System gets busy CPU has to do more work . Performance can slow down overall.
I think of it as students outside a classroom before a lecture. The teacher opens the door and everyone rushes in at once. The entry is not fast, they are getting in the way of each other and it's all crowded and slow. A similar problem can occur with computers . When there are many processes competing for one resource at a time .

12. Takeaways from India AI Impact Summit 2026.
- AI must not be just for big business and developed countries.
- Responsible AI needs to be fair, safe, transparent and should respect privacy.
- India aspires to be a leading nation for AI globally by building advanced artificial intelligence technologies and solutions for the world.
- Artificial intelligence is applicable to improve many sectors such as healthcare, education, agriculture, finance and public services.
- The challenges and opportunities associated with AI are international and require international cooperation.
- Future growth will require investment in AI infrastructure, including data centers and computers.
- It is important to teach AI skills to students and professionals, so they are prepared for jobs of the future.
- Ethical and security concerns must be taken in consideration at all stages of the development and deployment of standard artificial intelligence (AI) systems.

- Startups and research institutions are the main drivers of new AI solutions and economic growth.
- AI should not remain confined to the research and experiments .It should be used to solve the problems of the real world and to improve the quality of life of the people.

13.The Imitation Game

The Imitation Game taught me that intelligence alone is not enough, determination and patience are equally important. The film is about Alan Turing using his knowledge and not giving up when people doubted him. His work showed how the ideas of a single man can affect the world.
I learned to be a team player. Turing had great knowledge but could not have done his job without the help of his team. It taught me that working together beats working alone.